

Singular-Safe Port Compression and Finite-State Multiplicity Registers

Exact homology beyond direct-sum sharing, with a conditional
route to quasi-polynomial unknot recognition

Research continuation for the ProveIt project

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Abstract

Explicit chain-object multiplicity remains an asymptotic obstruction in the ProveIt Khovanov scanner. Literal direct-sum sharing does not describe coupling between copies. We develop an exact scalar-complex backend for a different representation: small template differentials repeated over exponentially many Boolean words, connected by a bounded number of rank-one channels, called ports. The port coordinates are given by layered weighted automata over $\mathbb{F}_2$, optionally restricted to an explicitly supplied deterministic language.

A singular-safe bordered-rank formula reduces the calculation to a matrix with at most $2r$ rows and columns, where r is the port count. Its use does not require an inverse of the port interaction matrix. Finite-state row spaces and parity pairings construct this matrix without enumerating the multiplicity domain. The same primitives check the full square-zero identity and minimize the port count exactly for the chosen base differential, without increasing register width. We prove a sharp rank-budget threshold for decision saturation and a tensor-amplification obstruction to automatic closure of the representation.

The resulting algorithm is polynomial in the explicit register presentation, not in the virtual chain dimension. A certified quasi-polynomial-time presentation producer would therefore yield a quasi-polynomial recognizer; no such producer for arbitrary diagrams is established here. The package contains proofs, standard-library Python, 48 unit tests, 4,352 exhaustive rank-update comparisons, 404 braid-closure cube audits, and both favorable and unfavorable paired measurements. Exponentially large abstract examples are not presented as knot benchmarks or as evidence of an unrestricted recognition bound.

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Reading guide. Sections 3–7 give the exact algebra, finite-state construction, validity test, and presentation-size bound. Sections 8 and 9 establish the sharp cancellation budget and a limitation on propagating scalar ports. For implementation and evidence, read Sections 11 and 12, then Appendix B. The constructive diagram-to-presentation question is isolated in Theorem 7.2 and the research agenda.

1 The bottleneck and the precise advance

1.1 What was inspected

The inspected ProveIt implementation stores the objects of each scanning stage in explicit lists, interns matching types, and uses sparse, prioritized unit cancellation. The complete default method still has an exponential general guarantee. Our source audit covers the project README, the beginning of `scan_fast.py`, and the maintained structural and radical-transfer sections; it is not a full checkout audit [8, 9, 10, 11]. The observed branch commit was `81a387aada56bb180649a98c21c078cb89efcfde`. The scanner blob was checked at that commit. Other inspected blobs and the exact scope of their retrieval are recorded separately in `provenance.json`.

Two existing ideas are particularly relevant. Component sharing compresses *literal direct sums*: equal differential components may be represented once with a multiplicity. Residue prediction determines how many matching copies survive scalar-unit cancellation, but not all surviving maps. The present work changes the multiplicity representation rather than merely accelerating another coefficient operation. It permits nonzero maps between copies, including examples whose entire differential support graph is connected.

This is a different contribution from the earlier twist, rank-two braid, extremal-window, graded-occupancy, and common-disk radical-transfer routes. It also has a different limitation: our verified algorithm operates on closed, scalar chain complexes. A closed homology rank is not a replacement for the relative chain-homotopy data of an open tangle.

1.2 Relation to the literature

Bar–Natan’s local simplification method provides the computational setting [1]. Kelomäki and Schütz show that the usual scanning and explicit-basis divide-and-conquer algorithms can require exponential time even on certain positive or alternating three-braids, while a different algorithm computes three-braid homology in polynomial time [2]. The distinction between an explicit basis and a succinct description is therefore substantive, not merely a programming preference.

The linear algebra below uses classical rank-factorization and bordered matrix operations. The register representation is a layered version of a weighted automaton; forward-space methods have established polynomial algorithms [5], and matrix-product representations also fit the tensor-train viewpoint [6]. We do not claim these ingredients, Lucas parity, or the elementary rank inequality as new discoveries. The contribution is their precise synthesis into a singular-safe, exact, proof-audited interface for coupled multiplicities, including optimal fixed-base port reduction, domain restriction, and explicit failure modes. Literature priority for this combined formulation has not been established.

Lackenby’s July 2026 hierarchy paper is a separate route. Section 9 explicitly leaves several operation costs and controlling parameters outside a complete running-time estimate [7]. The scalar-complex results here neither supply those missing geometric operations nor contradict the announced quasi-polynomial program motivating the repository.

1.3 Status of the claims

Claim	Scope	Status
Small bordered core	Arbitrary binary low-rank matrix updates, including singular interaction	Proved and tested
Succinct homology	Valid port-register complexes with explicit finite-state descriptions	Proved and implemented
Minimum port count	Fixed base differential and homogeneous scalar ports	Proved and implemented
Knot interpretation	A separate certified association with closed Khovanov homology	Conditional on provenance
General quasi-polynomial recognition	Arbitrary ordinary knot diagrams	Not established by this work

The measurements support a potential kernel improvement, not a default-backend change. A producer that first expands a small cube and then applies our interface is much slower than direct rank computation on that cube. The useful regime is one in which the producer avoids expansion in the first place.

2 Scalar complexes, templates, and ports

All coefficient arithmetic is over $\mathbb{F}_2$. We use cochain conventions: a differential raises homological degree by one. Quantum grading is not needed for the total-rank decision studied here and is not computed by the prototype.

Definition 2.1 (Port-register presentation). For each block ν , choose a finite graded space T_ν of dimension d_ν , a degree-one matrix A_ν satisfying $A_\nu^2 = 0$, and a domain $\Omega_\nu \subseteq \{0, 1\}^{m_\nu}$. In the basic model the domain is all words. The language-restricted model supplies an explicit layered deterministic automaton accepting exactly Ω_ν .

Set

$$C = \bigoplus_{\nu} T_{\nu} \otimes \mathbb{F}_2^{\Omega_{\nu}}, \quad A = \bigoplus_{\nu} A_{\nu} \otimes I_{|\Omega_{\nu}|}, \quad D = A + UV, \quad (2.1)$$

where U has r columns and V has r rows. Port j has a prescribed integer ℓ_j : its row in V is supported in degree ℓ_j , and its column in U is supported in degree $\ell_j + 1$. At each block and word, a weighted register simultaneously specifies the $d_\nu r$ coordinates of U and the $d_\nu r$ coordinates of V . Validity requires $D^2 = 0$.

Thus a port is a homogeneous rank-one channel, not a topological boundary circle or a cobordism generator. Several ports can be linearly dependent, and cancellation can make $UV = 0$ even when neither factor is zero. No positivity is assumed for differential entries. Multiplicity counts, by contrast, are ordinary nonnegative integers and are never reduced modulo two.

Write

$$N = \sum_{\nu} d_{\nu} |\Omega_{\nu}|, \quad q_{\nu} = \text{rank } A_{\nu}, \quad q = \sum_{\nu} q_{\nu} |\Omega_{\nu}|, \quad \beta_0 = \sum_{\nu} (d_{\nu} - 2q_{\nu}) |\Omega_{\nu}|. \quad (2.2)$$

The potentially enormous integers N, q, β_0 have small binary representations relative to the explicitly listed register lengths. The main target is

$$\beta(D) = \dim_{\mathbb{F}_2} H(C, D) = N - 2 \text{rank } D. \quad (2.3)$$

The last equality uses $D^2 = 0$, hence $\text{im } D \subseteq \ker D$. Applying it to an arbitrary matrix without the square-zero hypothesis would not compute homology.

2.1 The knot criterion and its provenance requirement

For a validated one-component classical knot diagram, total unreduced Khovanov rank two over $\mathbb{F}_2$ detects the unknot. Here is the precise chain of implications used by the project. Shumakovitch's splitting over characteristic two gives twice the reduced total rank [4, Corollary 3.2.C]. If the reduced rank over $\mathbb{F}_2$ is one, the universal coefficient theorem bounds its rational rank by one. That rational rank is nonzero: the reduced Jones Euler characteristic has value one at its standard identity specialization for a knot. Kronheimer–Mrowka's detection theorem then implies that the knot is trivial [3]. The unknot itself has unreduced rank two.

For completeness, the characteristic-two splitting comes from the marked point map X and the dot-derivative map ν , which commute with the chain differential and satisfy

$$X^2 = \nu^2 = 0, \quad X\nu + \nu X = I.$$

These identities split the complex into two isomorphic copies of the reduced complex, with quantum shifts ignored. They explain why the factor two is a mathematical feature, not a numerical convention.

Remark 2.2 (An abstract rank-two complex is not a knot certificate). The presentation (2.1) alone says nothing about which diagram, if any, it describes. The implementation deliberately emits `topological_verdict: null`. A trusted diagram-to-complex constructor, or a separately verified equivalence certificate, is needed before the rank criterion can be used as a topological verdict. Link inputs also require their own interpretation; the one-component promise is not dropped.

3 A bordered-rank formula without an inverse assumption

Lemma 3.1 (Reflexive inverse from a rank minor). *Let $A \in \mathbb{F}_2^{a \times b}$ have rank q . There is a matrix $h \in \mathbb{F}_2^{b \times a}$ satisfying*

$$AhA = A, \quad hAh = h. \quad (3.1)$$

It is constructible by elementary elimination. If I, J select an invertible $q \times q$ rank minor $M = A[I, J]$, then $h = E_J M^{-1} E_I^T$ is such a matrix.

Proof. The selected columns span the column space of A , and the selected rows of those columns are independent. Consequently $A = A[:, J]M^{-1}A[I, :]$. Substitution gives $AhA = A$. The second identity follows by replacing the middle $E_I^T A E_J$ with M . The empty rank-zero minor gives $h = 0$, including zero-dimensional cases. This is a reflexive generalized inverse, not a positive-definite or Moore–Penrose construction. $\square$

Theorem 3.2 (Singular-safe port core). *Let $A \in \mathbb{F}_2^{a \times b}$, $U \in \mathbb{F}_2^{a \times r}$, and $V \in \mathbb{F}_2^{r \times b}$. Choose h as in Lemma 3.1, and define*

$$U_0 = (I - Ah)U, \quad V_0 = V(I - hA), \quad E = I_r + VhU. \quad (3.2)$$

Let $\overline{U}$ consist of independent actual rows spanning the row space of U_0 , and let $\overline{V}$ consist of independent actual columns spanning the column space of V_0 . If these ranks are s, t , put

$$K = \begin{pmatrix} 0_{s \times t} & \overline{U} \\ \overline{V} & E \end{pmatrix}. \quad (3.3)$$

Then

$$\boxed{\text{rank}(A + UV) = \text{rank } A + \text{rank } K - r.} \quad (3.4)$$

The core has $(s + r) \times (t + r)$ entries, with $s, t \leq r$. The matrix E may be singular or zero. No inverse of E is used.

Proof. Consider the bordered matrix

$$B = \begin{pmatrix} A & U \\ V & I_r \end{pmatrix}.$$

Subtracting U times the bottom block row from the top block row shows that $\text{rank } B = r + \text{rank}(A + UV)$; subtraction equals addition in $\mathbb{F}_2$.

The reflexive identities decompose the domain of A as $\text{im } h \oplus \ker A$ and its codomain as $\text{im } A \oplus \ker h$. On the first summands A and h are inverse isomorphisms. Choose compatible bases so that both are the appropriate rectangular versions of $\text{diag}(I_q, 0)$. In these bases write $U = (U_1, U_2)^T$ and $V = (V_1, V_2)$. Eliminating the I_q block from B leaves

$$\begin{pmatrix} 0 & U_2 \\ V_2 & I_r + V_1 U_1 \end{pmatrix},$$

and therefore adds exactly q to the rank. In the same bases U_0 has rows $(0, U_2)^T$, V_0 has columns $(0, V_2)$, and $VhU = V_1 U_1$.

It remains to compress the two zero-corner directions. An invertible row operation on the top row block puts any chosen row basis of U_0 first and makes all its other rows zero. Because the upper-left block is zero, this does not change E or V_0 . The analogous column operation on the left column block keeps a column basis of V_0 and zeroes the others, without changing E or the retained rows of U_0 . Deleting the resulting zero rows and columns leaves precisely (3.3), up to independent basis changes in its top and left blocks. Thus $\text{rank } B = q + \text{rank } K$. Equating the two expressions for $\text{rank } B$ proves the result. $\square$

Example 3.3 (The smallest resonance). For $A = U = V = (1)$ one has $A + UV = 0$ and $E = 0$. An inverse-based update is undefined. Theorem 3.2 instead has $s = t = 0$, $K = (0)$, and $1 + 0 - 1 = 0$ as required. This is the essential singular case, not an exception to be handled by returning an inconclusive result.

Corollary 3.4 (Exact homology from the core). *For a valid presentation (2.1), take $h = \bigoplus_\nu h_\nu \otimes I$, using a reflexive inverse of every template. Then*

$$\boxed{\beta(D) = \beta_0 + 2r - 2\text{rank } K.} \quad (3.5)$$

Proof. Substitute (3.4) into (2.3) and use $\beta_0 = N - 2q$. $\square$

The formula is independent of which rank minor produces h , although the core matrix itself need not be canonical. It is also valid with redundant ports and with zero-dimensional blocks. These edge cases are included in the tests, rather than being excluded by an informal genericity assumption.

3.1 Homological degrees can be retained

The implementation reports total rank. The same theorem computes a graded profile by applying it separately to each rectangular differential $D_i : C^i \rightarrow C^{i+1}$, with the ports of degree i . If $R_i = \text{rank } D_i$ and $N_i = \dim C^i$, then

$$\dim H^i(C, D) = N_i - R_{i-1} - R_i.$$

The total port count is the sum of these degree-group counts, so summing the corresponding cubic bounds remains polynomial. This is an extension of the proof, not a claim that the shipped command-line interface already outputs quantum grading, integral torsion, representatives, or relative chain maps.

4 Finite-state multiplicity registers

Definition 4.1 (Layered weighted register). A length- m register with Q outputs consists of positive integers $b_0, \dots, b_m$, an initial row $\alpha \in \mathbb{F}_2^{b_0}$, two matrices $T_j^0, T_j^1 \in \mathbb{F}_2^{b_{j-1} \times b_j}$ for every layer, and a terminal matrix $O \in \mathbb{F}_2^{b_m \times Q}$. It evaluates to

$$F(x_1, \dots, x_m) = \alpha T_1^{x_1} \dots T_m^{x_m} O. \quad (4.1)$$

Its width is $B = \max_j b_j$. All layers are explicitly listed; m is not an exponentially large repetition count encoded in binary.

For block ν the output count is $Q = 2d_\nu r$. The first $d_\nu r$ bits encode the rows of U at one word, and the next $d_\nu r$ bits encode the columns of V at that word. This simultaneous encoding permits shared finite-state structure among different port coordinates. It is not a requirement to build Q separate automata.

A product of literals is represented by a diagonal register: one state per product term, each transition retaining or deleting that state according to the selected literal value. In particular, a point indicator needs one term, not the exponentially long expansion that an unfortunate polynomial basis might produce. General dense transition matrices are also allowed.

Theorem 4.2 (Reachable rows with actual witnesses). *The span of the 2^m evaluation rows $F(x)$ has a basis consisting of at most B actual evaluations. Such a basis and its witnessing words can be computed using $O(mB^3 + B^2Q)$ binary field operations. The witness bookkeeping requires only $O(mB)$ predecessor records.*

Proof. At layer zero retain α if it is nonzero. Suppose rows $w_1, \dots, w_s$, each realized by a prefix, span all reachable rows at layer $j - 1$. Every reachable row at layer j lies in the span of

$$w_i T_j^0, \quad w_i T_j^1 \quad (1 \leq i \leq s).$$

Conversely these candidates are themselves realized by prefixes. Choose an independent subset of the actual candidates, not merely arbitrary row-reduced substitutes. There are at most $b_j \leq B$ selected rows. Induction proves both the spanning property and the existence of actual prefix witnesses. At the terminal layer multiply by O and select an independent subset again.

There are at most $2B$ candidates per layer; multiplication and elementary elimination cost $O(B^3)$ field operations. Terminal multiplication and selection cost $O(B^2Q)$. Store each new witness as a predecessor pointer and one appended bit. At most B records are retained per layer. Recovering the at most B final words then costs $O(mB)$ steps. Copying whole prefixes at every layer would instead introduce an avoidable quadratic factor in m . $\square$

This is a specialization of the established forward-space idea for weighted automata [5]. Keeping actual witnesses makes the implementation and its audit particularly transparent: each claimed basis row is checked by replaying an ordinary binary word.

Theorem 4.3 (Exact parity pairings). *Let $w(x) = \alpha T_1^{x_1} \cdots T_m^{x_m}$ be the terminal state row, and define*

$$W_0 = \alpha^T \alpha, \quad W_j = (T_j^0)^T W_{j-1} T_j^0 + (T_j^1)^T W_{j-1} T_j^1. \quad (4.2)$$

Then $W_m = \sum_x w(x)^T w(x)$. For terminal coefficient columns a, b ,

$$\sum_x (w(x)a)(w(x)b) = a^T W_m b. \quad (4.3)$$

All sums here are in $\mathbb{F}_2$.

Proof. Partition the length- j words by their last symbol and apply induction to the outer product of their state rows. The scalar formula follows by multiplying by a^T and b . Dense evaluation of the recurrence costs $O(mB^3)$ field operations. $\square$

Remark 4.4 (A Gram matrix is not a rank test in characteristic two). For the nonzero constant function on $\{0, 1\}^m$, $m \geq 1$, the evaluation matrix has rank one while its self-pairing is $2^m = 0$ in $\mathbb{F}_2$. Thus a Gram matrix can vanish on a nonzero row space. The implementation uses reachable spaces for ranks and the recurrence (4.2) only for contractions. It never substitutes one task for the other.

4.1 How the core is computed without expansion

Apply Theorem 4.2 to the simultaneous output register. For each retained evaluation and each template coordinate, compute the corresponding row of $(I - A_\nu h_\nu)U$ and column of $V(I - h_\nu A_\nu)$. The union of these finite lists spans all rows and columns needed for $\bar{U}, \bar{V}$. Indeed, each coordinate extraction and template matrix action is linear in the simultaneous evaluation row.

The entries of VhU are sums, over templates and words, of products of a V coordinate and an hU coordinate. Theorem 4.3 computes these sums by small terminal coefficient matrices and W_m . The implementation does *not* build a full $Q \times Q$ output Gram matrix. It contracts the $B \times B$ state matrix directly against $B \times r$ port coefficient arrays, one template coordinate at a time.

4.2 Elementary closure properties and their cost

A fixed linear transformation of the output functions changes O only; it leaves every transition matrix and the width unchanged. A sum of two register functions has a direct-sum representation of width at most $B_1 + B_2$. Their pointwise product has a tensor-product representation of width at most $B_1 B_2$. These identities follow by distributing matrix products or by the elementary identity $(aA \otimes bB) = (a \otimes b)(A \otimes B)$.

These are exact operations over $\mathbb{F}_2$, with no numerical tolerance or singular-value truncation. They also expose a danger: repeated pointwise products can multiply widths. Closure under an operation is not a polynomial bound on the size after a long sequence of such operations.

5 Regular-language multiplicities and exact cardinalities

Boolean multiplicity domains are useful but need not include all words. For example, a fixed homological layer may correspond to words of a fixed Hamming weight. We allow an explicitly

supplied *deterministic* layered language automaton, with at most L states per layer, a single initial state, and designated accepting terminal states.

The number $a = |\Omega|$ is obtained by dynamic programming with integer counts: initialize the initial-state count to one, and for each symbol add each state's count to the count of its specified successor. There are $O(mL)$ additions of integers with at most $m + 1$ bits. Determinism matters: counting accepting paths of an arbitrary nondeterministic presentation would not in general count accepted words.

Proposition 5.1 (Language restriction without coordinate enumeration). *A width- B output register can be masked by the indicator of a supplied width- L deterministic language using a width-at-most- BL register. The core in Theorem 3.2, its validity checks, and the port ranks can then be computed over the accepted domain without enumerating accepted or rejected words. Exact dimension bookkeeping uses $|\Omega|$, not its parity.*

Proof. Take the tensor product of the weighted register with the deterministic state-transition representation of the language. At the terminal layer set an output row to zero unless the language state accepts. The value on a word is precisely $\mathbf{1}_\Omega(x)F(x)$. Hence the evaluation row spaces are those from accepted words, together with extra zero rows. The parity pairings are also correct because $\mathbf{1}_\Omega(x)^2 = \mathbf{1}_\Omega(x)$ in $\mathbb{F}_2$.

To see the dimension statement transparently, extend the masked model to all Boolean words. Its differential is a direct sum of the desired accepted-domain complex and

$$(T, A_T)^{\oplus(2^m - |\Omega|)}$$

on the rejected domain. No port enters or leaves the latter summand. Consequently its dimension, differential rank, and homology dimension are respectively

$$d(2^m - |\Omega|), \quad q_T(2^m - |\Omega|), \quad (d - 2q_T)(2^m - |\Omega|).$$

Subtract these *known exact integers* from the full-word calculation. The small core is unchanged, since all rejected-domain port coordinates are zero. This is also the implementation used by the optional language wrapper. The argument applies blockwise even when the accepted blocks are coupled to each other. $\square$

Remark 5.2 (Do not saturate before removing the known complement). The full-word padded model can have enormous homology entirely supported on rejected words. A capped value cannot be used for the exact subtraction in Proposition 5.1. The implementation first removes the known uncoupled complement using arbitrary-precision integers, and only then forms decision caps for the actual domain. A test with a zero base, 2^{40} padded copies, and only one accepted word has actual homology zero despite the enormous padded complementary homology.

The fixed-weight language $|x| = k$ has a deterministic counter with $k + 2$ states: counts $0, \dots, k$ and a dead overflow state. Its accepted cardinality is $\binom{m}{k}$. Thus the representation supports binomial, not just power-of-two, multiplicities at polynomial descriptor cost. No unproved statement about Khovanov diagrams having such a representation is implicit in this observation.

6 Checking the full differential and minimizing ports

6.1 A symbolic square-zero check

It is insufficient to check the template differentials and assume their coupling remains a complex. The following test evaluates all of D^2 , not a sample of its entries.

Theorem 6.1 (Finite-state validity certificate). *Assume $A^2 = 0$, and compute $M = VU$. Then*

$$D^2 = \begin{bmatrix} AU + UM & U \end{bmatrix} \begin{bmatrix} V \\ VA \end{bmatrix}. \quad (6.1)$$

The condition $D^2 = 0$ is equivalent to every pairing between a basis of the row space of the first factor and a basis of the column space of the second factor being zero. Both bases have size at most $2r$ and are constructible from the register evaluations without expanding C . Homogeneity of all ports is checkable by the same evaluations.

Proof. Expanding $(A + UV)^2$ and using $A^2 = 0$ gives $AUV + UVA + U(VU)V$, which is (6.1). Each entry of a product matrix is a pairing of a row of the first factor with a column of the second. Bilinearity makes vanishing on spanning bases equivalent to vanishing on every such pair.

The parity-pairing algorithm computes M exactly. At each template and word, $AU + UM$ and VA are linear expressions in the simultaneous register output after M is known. The reachable-row theorem therefore provides spanning lists for all required rows and columns. Ordinary elimination reduces them to at most $2r$ each.

A forbidden degree coordinate is likewise a fixed linear functional of the simultaneous output. It vanishes on every word exactly when it vanishes on a spanning set of evaluations. Checking these coordinates therefore certifies port homogeneity, including functions that vanish only after cancellation in the register. With language restrictions, use the masked evaluations and the actual accepted domain. $\square$

This check establishes an algebraic chain complex. It does not establish that the complex comes from a planar diagram, nor that its coefficients encode valid cobordisms before closure. Such geometry remains the responsibility of the producer.

6.2 The minimum number of ports for a fixed base

Redundant ports waste cubic core work and weaken rank-budget obstructions. Their redundancy can be removed exactly, without modifying register widths.

Theorem 6.2 (Exact port minimization). *For a fixed base differential A , the smallest number of homogeneous scalar ports in a factorization of $D - A$ equals*

$$r_{\min} = \text{rank}(D - A). \quad (6.2)$$

For a port-register presentation this number and a factorization attaining it are constructible in polynomial time in the presentation. Every register retains its length, transition matrices, and widths; only its terminal output coefficients change. This does not optimize the base A .

Proof. First work within one port-degree group and write the update as UV with k ports. Choose an actual row basis $P \in \mathbb{F}_2^{a \times k}$ of U and an actual column basis $Q \in \mathbb{F}_2^{k \times b}$ of V . These are computed using reachable register evaluations. Choose a right inverse P^+ and a left inverse Q^+ using rank minors. Define

$$L = UP^+, \quad R = Q^+V.$$

Because every row of U is in the row space of P , and every column of V is in the column space of Q , one has $U = LP$ and $V = QR$. Furthermore, the actual rows of U selected for P give an identity submatrix in L ; the actual columns selected for Q give one in R . Hence L has full column rank and R has full row rank.

It follows that

$$\text{rank}(UV) = \text{rank}(L(PQ)R) = \text{rank}(PQ).$$

Factor the small matrix $PQ = ST$ using exactly $r' = \text{rank}(PQ)$ columns in S . Then

$$U' = UP^+S, \quad V' = TQ^+V \quad (6.3)$$

satisfy $U'V' = UV$ and use r' ports. Every entry of U' and V' is a fixed linear combination of original port-coordinate functions. Thus (6.3) changes only terminal outputs of their registers.

Repeat separately for each degree. Distinct groups map distinct source degrees to distinct target degrees, so the ranks of these blocks add. The result has $\text{rank}(D - A)$ homogeneous ports. Conversely any sum of s rank-one channels has rank at most s , proving minimality. All matrices used to form the port transformations have dimensions bounded by the original port count. No basis vector of the virtual space is enumerated. $\square$

Example 6.3 (Duplicate channels can disappear completely). Two identical columns of U paired with two identical rows of V contribute twice the same rank-one matrix and cancel over $\mathbb{F}_2$. Both factors may be nonzero, but the optimal port count is zero. The prototype checks this on a length-40 register. Minimizing $\text{rank } U$ and $\text{rank } V$ separately would not detect every such cancellation: the small interaction PQ is essential. Nor can one substitute $\text{rank}(VU)$ for $\text{rank}(UV)$: with an even number M of coordinates, $U = \mathbf{1}$ and $V = \mathbf{1}^T$ give $VU = 0$ but $UV = J_M$ has rank one.

The theorem is a useful preprocessing operation because it improves both the core dimension and the sharp rank budget. It is not an algorithm for finding a low-rank perturbation of an arbitrary differential: the quality of the *chosen base* is a distinct and harder issue.

7 The main presentation-size complexity theorem

Let the number of template blocks be b . For block ν , let d_ν be the template dimension, m_ν the explicitly listed register length, and B_ν the maximum *effective* width after any language mask. The unmasked register width times the deterministic language width bounds B_ν . Define

$$\begin{aligned} \Psi_\nu &= d_\nu^3 + (m_\nu + 1)B_\nu^3 + d_\nu^2 B_\nu r \\ &\quad + d_\nu B_\nu^2 r + d_\nu B_\nu r^2. \end{aligned} \quad (7.1)$$

Theorem 7.1 (Polynomial-time homology of supplied presentations). *From an explicit port-register presentation one can deterministically check the template and port degree conditions, check $D^2 = 0$, and, when valid, compute $\beta(D)$ exactly without enumerating any Ω_ν . The binary field-operation cost is*

$$O\left(\sum_{\nu=1}^b \Psi_\nu + r^3\right). \quad (7.2)$$

The algorithm uses polynomial space in the descriptor. Integer cardinality, index, and dimension arithmetic has polynomial bit cost in the explicit input length. For supplied width- L_ν language automata, a conservative bound for the cardinality additions is $O(\sum_\nu (m_\nu + 1)^2 L_\nu)$ bit operations. The optional exact minimization of Theorem 6.2 preserves this polynomial bound. No factor 2^{m_ν} occurs in the field-operation bound.

Proof. For each template, check $A_\nu^2 = 0$ and compute a reflexive inverse in $O(d_\nu^3)$ field operations. Run the reachable-space and parity-pairing algorithms. Their per-block work is bounded by $O((m_\nu + 1)B_\nu^3 + d_\nu B_\nu^2 r)$, since the simultaneous output count is $2d_\nu r$.

Multiplying template matrices into the port coefficient arrays costs $O(d_\nu^2 B_\nu r)$. The contractions producing VU and VhU cost $O(d_\nu B_\nu^2 r + d_\nu B_\nu r^2)$. Acting on the at most B_ν retained evaluations to form quotient and validity rows has the same bounds. Their total number is $O(\sum_\nu d_\nu B_\nu)$; each row has at most $2r$ bits, so eliminating those lists costs $O(\sum_\nu d_\nu B_\nu r^2)$. Checking all pairings of the two small validity bases and reducing the core costs $O(r^3)$.

Equations (2.2) and (3.5) finish the calculation. Full-word dimensions require only binary shifts. A language count uses $O(m_\nu L_\nu)$ additions with $O(m_\nu)$ -bit operands, which gives the stated conservative cardinality cost. Dimensions and ranks have at most

$$O\left(\max_\nu m_\nu + \log\left(1 + \sum_\nu d_\nu\right)\right)$$

bits. All explicitly supplied degree labels and matrix indices are also accounted for by the input bit length.

The stored transition matrices, template matrices, terminal coefficient arrays, small cores, and reachable witnesses occupy polynomial space. In particular, witness construction uses shared predecessor records, not arrays indexed by all words. Port minimization adds row-space extraction and matrices of size at most r , together with fixed linear output transformations; its cost fits the same polynomial estimate. $\square$

The estimate is conservative elementary linear algebra, not an assertion that Python takes one machine instruction per field operation. Arbitrary-precision bit rows often make the implementation substantially faster than that elementary estimate. Conversely the measured timings cannot be used to replace the proof with a fitted exponent.

7.1 A concrete algorithm contract

A producer may provide a compressed presentation, but the consumer does not trust cached ranks or an asserted square-zero identity. It performs these steps:

1. Validate template shapes and degrees, build any deterministic language mask, and compute the accepted multiplicities as integers.
2. Compute template inverses, finite-state evaluation bases, parity pairings, and the complete small square-zero check.
3. Optionally minimize the homogeneous ports while preserving the base.
4. Construct $\bar{U}, \bar{V}, E, K$, compute rank K , and evaluate (3.5); remove any known padded complementary summands before applying a decision cap.
5. Return the algebraic result and its replay data. Apply a topological verdict only in a caller with independent diagram-to-complex provenance.

The shipped `verify` command recomputes this contract and compares its canonical output with the recorded certificate. It shares the solver implementation and is therefore a *replay checker*, not a separately implemented proof checker. The independent small-matrix and crossing-cube audits provide different checks, but are finite tests rather than formal verification.

7.2 What yields a quasi-polynomial recognition bound

Theorem 7.2 (Constructive, conditional recognition theorem). *Let $n \geq 2$ be the encoded input length of a validated knot diagram. Suppose a proved, constructive procedure produces and certifies a presentation whose homology is that of its closed unreduced Khovanov complex over $\mathbb{F}_2$. Suppose its construction and provenance-verification cost is $n^{O(1)}2^{O(\sigma(n))}$, the number of blocks and all register lengths are polynomial in n , and*

$$d_\nu, B_\nu, r \leq n^{O(1)}2^{O(\sigma(n))}$$

with the integer label lengths polynomially bounded as well. Then the combined recognition procedure has bit complexity

$$n^{O(1)}2^{O(\sigma(n))}. \quad (7.3)$$

In particular $\sigma(n) = O(\log^2 n)$ gives $n^{O(\log n)}$ time.

Proof. Every term of (7.2), and the additional integer bookkeeping, has the stated bound after substitution. Add the assumed production and verification cost. The rank-two criterion from Section 2 then decides the input knot. $\square$

This theorem is useful as an implementation target because the consumer and all of its costs are now explicit. It is not a substitute for the producer hypothesis. Merely postulating a small presentation, finding one by exponential search, constructing one after an exponential scan, or assuming a short certificate without an efficient verifier does not meet the theorem. The largest unresolved task is controlling the construction and maintenance of this representation on ordinary diagrams.

8 Sharp rank budgets and safe decision saturation

Theorem 8.1 (Sharp perturbation bound). *For valid square-zero matrices A, D on the same finite graded space, with $D - A = UV$ and r ports,*

$$|\beta(D) - \beta(A)| \leq 2 \operatorname{rank}(UV) \leq 2r. \quad (8.1)$$

Consequently $\beta_0 > 2r + 2$ is a rank-two obstruction. For the rank-two query it is sufficient to retain the base count as

$$\min(\beta_0, 2r + 3) \quad (8.2)$$

along with the core data. After port minimization the optimal fixed-base budget is obtained by replacing r with $r_{\min}$.

Proof. The inequalities $\operatorname{rank} D \leq \operatorname{rank} A + \operatorname{rank}(UV)$ and $\operatorname{rank} A \leq \operatorname{rank} D + \operatorname{rank}(UV)$ give the rank bound. Equation (2.3) turns it into (8.1). If $\beta_0 > 2r + 2$, the final homology dimension exceeds two. Otherwise the capped value (8.2) retains β_0 exactly, so (3.5) is still an exact test for equality to two. $\square$

Proposition 8.2 (Sharpness and the failure of a fixed cap of three). *The obstruction threshold $2r + 2$ cannot be lowered using the rank budget alone. The direct-sum cap $\min(3, \beta_0)$ is insufficient for arbitrary coupled presentations, even with one port.*

Proof. Take $A = 0$ on a two-term space of total dimension $2r + 2$, with $r + 1$ generators in each degree. Let D contain r disjoint identity arrows. Then $D^2 = 0$, $\text{rank } D = r$, and $\beta(D) = 2$. Thus a base dimension of $2r + 2$ can still lead to rank two. Removing the two isolated generators shows the full downward change $2r$ is attained. Conversely, starting with r contractible pairs and cancelling all their arrows by an equal perturbation attains the upward change $2r$.

For the fixed-cap failure, take $r = 1$. A zero base of dimension four with a rank-one two-term differential has final homology dimension two. Adding two isolated generators gives base dimension six and final homology dimension four. In both examples the nontrivial core is the same 2×2 matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

of rank two, and both base dimensions would be capped to three. The capped base and unchanged core would no longer distinguish the cases. $\square$

This is an extension-specific warning, not a defect in the maintained literal-direct-sum sharing method. Nonnegative multiplicity saturation is sound when no differential connects the summands. Coupling introduces new cancellations, so the available cancellation rank must enter the threshold.

Corollary 8.3 (Necessary near-contractibility of a useful unknot base). *If a fixed-base presentation represents a knot of total rank two, then*

$$r_{\min} \geq \frac{\beta_0 - 2}{2}.$$

In particular, a huge zero-differential base cannot be repaired into an unknot complex using only a small number of scalar ports.

This gives a practical design constraint: a promising producer must identify most contractible structure *before* encoding the remaining couplings. Assigning every expanded generator to a zero template and hoping that a few global channels will reconstruct the differential is not a viable general strategy.

8.1 Future updates need a future budget

Suppose, on a fixed ambient space, a current valid differential may later receive updates whose ranks sum to at most R , and the final differential is again valid. The same proof gives a possible homology change of at most $2R$. A rank-two obstruction must therefore reserve that whole remaining budget. A bound using only channels already processed would be unsound. This observation does not automatically apply to future crossing additions, which usually change the ambient complex and require relative gluing data.

9 An obstruction to automatic scalar-port closure

Finite-state compression solves evaluation and rank extraction for the specified model. It cannot by itself make the model closed under every operation that occurs in a topological scanner.

Theorem 9.1 (Tensor amplification of the minimum port count). *Fix a base A and an update $\Delta = D - A$ of rank $s > 0$. If both the base and differential are tensored with an identity on a*

q -dimensional multiplicity space, then the minimum scalar-port count for that fixed new base is exactly

$$\text{rank}(\Delta \otimes I_q) = sq. \quad (9.1)$$

No choice of scalar register-coordinate functions can represent this same fixed-base update with fewer than sq ports.

Proof. Choose bases reducing Δ to an identity block of size s and a zero block. Tensoring these invertible basis changes with I_q reduces $\Delta \otimes I_q$ to an identity block of size sq and a zero block. The lower bound on any port factorization follows from rank. A rank factorization realizes equality, so the bound is exact. $\square$

Even appending t completely passive binary multiplicity factors can therefore change a nonzero fixed-base port requirement from s to $s2^t$. This is a genuine algebraic obstruction, not a poor choice of weighted-automaton basis. It rules out the argument that a compact register description alone guarantees small port count under arbitrary tensor extension.

There are possible responses, but none is free: absorb the replicated update into a newly chosen template; allow structured matrix-valued ports; use a recursively factored coupling description; or prove that the particular geometric operation has smaller effective rank after additional certified cancellations. Each option changes the construction problem and needs its own size theorem. The existing structured-tensor and radical methods may help with such absorption, but their compatibility with this closed scalar interface is not asserted without proof.

10 Explicit families and what they demonstrate

10.1 Connected support with one port and rank-two homology

Let $M = 2^m$, $m \geq 2$, and consider the two-term complex

$$\mathbb{F}_2^M \xrightarrow{I_M + \mathbf{1}(e_0 + e_1 + e_2)^T} \mathbb{F}_2^M. \quad (10.1)$$

Use the two-dimensional template with one identity arrow, so the base is contractible. The column U is the constant function and the row V is the sum of three point indicators. A diagonal register of width four encodes them for every m .

Proposition 10.1 (An exponentially large connected example). *The differential in (10.1) has rank $M - 1$, and its total homology dimension is two. Its undirected differential support graph is connected. It has a one-port presentation with a 1×1 zero core, independent of m .*

Proof. Put $v = e_0 + e_1 + e_2$. Then $v^T \mathbf{1} = 3 = 1$ in $\mathbb{F}_2$. If $(I + \mathbf{1}v^T)z = 0$, then $z = \mathbf{1}(v^T z)$, so the kernel is contained in the span of $\mathbf{1}$. That vector is in the kernel, giving nullity one and rank $M - 1$. The two-term cokernel has the same dimension.

For connectedness, the source vertices $0, 1, 2$ are each adjacent to all targets except the corresponding target of the same index. The subgraph on these six vertices is a six-cycle. Every target with index at least three is adjacent to all three of these source vertices. Every remaining source has its identity edge to its own target. Thus all vertices lie in one component.

The template inverse gives $U_0 = V_0 = 0$ and $E = 1 + v^T \mathbf{1} = 0$. The core consists of this scalar, so (3.5) gives $0 + 2 - 0 = 2$. $\square$

This is a strict representation distinction from sharing literal connected components *of the supplied matrix*: the input has no nontrivial differential-component decomposition to share. It is not a lower bound against every algorithm that may first change basis or perform cancellation. In particular, explicit scalar elimination will reduce this example once it has allocated or otherwise accessed the relevant matrix entries. The advantage demonstrated here is that the register algorithm need not allocate its virtual generators at all.

For $m = 40$, the represented chain space has $2^{41} = 2,199,023,255,552$ generators. The program also checks $m = 200$ and $m = 1000$ using the same one-port construction. These are *abstract* chain complexes. No knot diagrams with these prescribed presentations were constructed in this work.

10.2 Dense parity-sensitive coupling

Replace v by $\mathbf{1}$ and restrict the multiplicity domain to words of Hamming weight k . Write $a = \binom{m}{k}$. The resulting nonzero block is

$$I_a + J_a, \quad J_a = \mathbf{1}\mathbf{1}^T. \quad (10.2)$$

Only one rank-one channel is needed. The unmasked output register has width one; the deterministic counter mask has width $k + 2$.

Proposition 10.2 (Binomial resonance). *The two-term complex with differential (10.2) has total homology dimension*

$$\beta = \begin{cases} 2, & \binom{m}{k} \text{ is odd,} \\ 0, & \binom{m}{k} \text{ is even.} \end{cases} \quad (10.3)$$

The odd case occurs exactly when every nonzero binary digit of k is also a nonzero binary digit of m .

Proof. Here $E = 1 + a$ in $\mathbb{F}_2$. If a is even, then $J_a^2 = aJ_a = 0$, so $(I_a + J_a)^2 = I_a$ and the differential is an isomorphism. If a is odd, the same kernel argument as above gives nullity one. The case $a = 0$ has zero-dimensional chain groups and belongs to the even case.

For the parity criterion, write $m = \sum_j m_j 2^j$. In $\mathbb{F}_2[t]$,

$$(1 + t)^m = \prod_{j:m_j=1} (1 + t^{2^j}).$$

Each exponent in the product is the sum of a unique subset of the indicated powers of two. The coefficient of t^k is therefore one exactly under the stated binary condition. This is the familiar binary Lucas criterion; it is used here as an independent closed-form check, not claimed as a new combinatorial theorem. $\square$

The packaged language examples have the following exact values:

(m, k)	Multiplicity a	Total dimension $2a$	β
(63, 31)	916312070471295267	1832624140942590534	2
(64, 32)	1832624140942590534	3665248281885181068	0

The first case has singular interaction and the second has invertible interaction. Both are processed by the same formula. Integer cardinality is essential for the virtual dimension, while parity is the correct arithmetic for VhU ; conflating these two roles would destroy the result.

10.3 Large surviving homology is also represented exactly

Take a zero base and the rank-one two-term differential $\mathbf{11}^T$ on M generators in each degree. Its homology dimension is $2M - 2$ for $M \geq 1$. The register algorithm outputs this integer without a basis of that size. This example tests the opposite regime from the contractible base: a small number of ports cannot remove the large base homology, exactly as Theorem 8.1 predicts.

11 Integration with the maintained recognizer

11.1 A deliberately narrow adapter

The module `portkh.explicit` contains an explicit audit bridge. It accepts a scalar, graded, square-zero matrix; chooses vertex-disjoint arrows as a square-zero base; factors the residual separately in every degree; and constructs a zero-length register. Equality of the resulting matrix with the input is checkable by expansion on small cases. This is a correctness bridge, not an asymptotic compression algorithm: the entire explicit matrix is already present and the template can be as large as that matrix.

The companion `from_closed_fastscan` adapter targets the inspected interned-list API. It refuses a nonempty frontier, non-scalar coefficients, inconsistent live counts, mixed closed matching identifiers, and edges targeting deleted objects. It was exercised with API-shaped fixtures and independent small cubes. It was *not* exercised against a complete upstream checkout, the maintained 277-test suite, or the production PD corpus. These distinctions are also recorded in the integration notes.

Running this adapter after full scalar elimination is generally redundant: the surviving closed complex already has zero differential. Running it before that elimination still cannot recover the cost of a prefix scan that has already expanded exponentially. A genuine improvement needs a producer that emits shared templates and coordinate registers earlier, with the required relative information preserved until closure.

11.2 Data that must not be discarded at an open frontier

Before closure, differential coefficients are typed cobordisms rather than ordinary scalars. Surviving matching multiplicities do not determine all maps; the maintained radical discussion already gives examples of this issue [11]. The present use of a generalized inverse over $\mathbb{F}_2$ does not bypass that category structure. In particular, it does not license scalar mixing of inequivalent boundary objects, nor transfer of a common-disk radical exponent to an uncertified arbitrary PD frontier.

A future relative adapter must state the object types of its ports, the coefficient algebra of its maps, and the continuation information carried by the compressed basis. A theorem about the rank of one closed complex does not imply equivalence under every future gluing operation.

11.3 An adoption gate based on structural cost, not optimism

The natural first integration is an opt-in diagnostic, not replacement of `scan_fast.py`. A producer should report template dimensions, raw and minimized port counts, effective register widths, accepted-word counts, estimated descriptor storage, and the cost of obtaining all of those data. It should preserve the current fallback state until the new operation has passed its checks.

For ordinary already-expanded matrices, the bridge should remain disabled by default: the measured controls below show pronounced overhead. For a native compressed producer, an admission test can compare a bound such as (7.2) with an explicit-allocation estimate. Such a test is a resource

policy, not a knot invariant. A declined conversion or exhausted budget must fall back or report unknown, never be interpreted as proof of nontriviality.

The supplied package does not modify the repository, enable a new default, or assert production compatibility beyond the inspected interface. The recommended integration directory is a new named research subdirectory, not an assumed unused numeric report identifier.

12 Validation and measured performance

12.1 What was actually run

The standard-library test suite passes 48 tests. Several tests include seeded loops: 1,000 random rectangular rank updates, 300 reflexive-inverse checks, 200 rank factorizations, 300 random finite-state reachability and pairing comparisons, 250 coupled two-term examples, 160 three-term validity trials, 100 exact port-minimization comparisons, and 70 language-restricted dense comparisons. Additional tests cover literal truth tables, long shared witnesses, malformed input, degree failures, empty dimensions, duplicate ports, unsafe saturation, large dimensions, CLI replay, and certificate tampering. These are not all independent mathematical tests of the same strength; the raw test log identifies exactly what each method checks.

The separate validation script exhausts all binary 2×2 bases with one or two ports. This gives 4,352 update cases, including 1,728 with a singular middle matrix. Every result agrees with a distinct low-pivot rank implementation. The unit suite additionally compares against a dense list-of-bits elimination routine, rather than relying exclusively on the production bit-row elimination.

An independent braid-closure cube constructor audits every signed 2-strand word of length at most five and every signed 3-strand word of length at most four: 404 distinct presentations, of which 210 have one component and 194 have more than one. Many presentations represent the same knot or link. The largest cube dimension is 246. Every cube is checked for $d^2 = 0$, converted through the explicit bridge, re-expanded for equality, and compared in total homology. Each also has a mirror homological-profile check. The mirror set overlaps the original set; it is not another 404 distinct knot types.

Known small examples and tests of braid relations, cancellation, and positive and negative stabilization supplement this enumeration. The cube oracle is independent of the register and port construction, but is itself a new small implementation rather than an external production knot package. These finite checks support the code; they do not replace the proofs or constitute formal verification.

The files `data/validation.json`, `data/unit-tests.txt`, `data/benchmarks.json`, and the example certificates preserve the actual results. The scripts contain fixed random seeds. No full upstream checkout validation or end-to-end production recognition benchmark was run in this session.

12.2 Paired timings and their interpretation

All reported timings use seven rounds with randomized order among a baseline, the port implementation, and a second identical baseline. The A/A ratio measures local timing variability. Small cases use batches of five calls. Imports, parsing, and disk I/O are excluded. Host and Python metadata, every order, and every raw duration are stored in the JSON, so the displayed ratios can be reconstructed. A paired ratio is the median of within-round baseline/treatment ratios, not necessarily the quotient of the two marginal median times.

For the connected abstract family, the baseline is intentionally favorable: it constructs only the nonzero $M \times M$ block from its closed formula, then performs bit-packed elimination. It does not allocate a $2M \times 2M$ total differential. The treatment analyzes an already prepared compact register, including degree and square-zero validation. Both therefore start from compact descriptions, but their construction and checking work is not identical; the exact timed tasks are recorded. The family also has a closed-form answer: this benchmark compares two representation paths, not the best possible specialized algorithm for this family.

Table 1: Connected abstract examples: prepared-register analysis versus construction and rank of the explicit nonzero block. Seven randomized paired rounds. Times are milliseconds.

m	Virtual dimension	Explicit	Register	Paired ratio	A/A
4	32	0.0021	0.0832	0.025	0.913
6	128	0.0071	0.1025	0.073	0.944
8	512	0.0281	0.1143	0.245	0.989
10	2,048	0.1325	0.1369	0.962	1.039
12	8,192	0.5673	0.1637	3.491	1.001
14	32,768	7.0811	0.2394	31.631	1.090

Table 2: Compressed-only connected examples. No explicit competitor was run at these sizes; no speedup ratio is inferred.

m	Virtual dimension	Width	Ports	Time (ms)	Candidate rows
40	2^{41}	4	1	0.4141	310
200	2^{201}	4	1	2.0672	1590
1000	2^{1001}	4	1	11.7555	7990

Table 3: Negative controls on prepared small knot cubes. The treatment includes the explicit-to-port producer. Times are milliseconds; a ratio below one is a slowdown.

Presentation	Dimension	Ports	Direct	Bridge	Paired ratio
Stabilized unknot	18	8	0.0032	0.2852	0.0113
Trefoil	30	15	0.0064	0.6077	0.0108
Figure-eight	66	36	0.0220	2.0978	0.0105
Relator unknot	90	49	0.0351	3.5818	0.0098

The crossover is visible rather than hidden. At $m = 4$ the register method is approximately 39.5 times slower. At $m = 10$ the median paired ratio is 0.962, close to the observed crossover. At $m = 12$ and $m = 14$ the paired ratios are 3.49 and 31.63. These are abstract-kernel observations on this host, not a fitted asymptotic law and not knot-recognition speedups. The largest A/A variation in these tiny measurements also argues against excessive precision in performance claims.

The compressed-only cases demonstrate that the implementation follows its representation-size theorem: length 40, 200, and 1000 examples return the proved rank-two homology while the virtual dimensions are 2^{41} , 2^{201} , and 2^{1001} . No explicit run was attempted at those sizes, and no hypothetical elapsed time or speedup is attributed to one.

The four small knot-cube controls make the limitation concrete. Their prepared-cube direct baseline is much faster than the combination of the explicit-to-port producer and the symbolic solver. Ratios around 0.010 correspond to roughly two orders of magnitude of overhead. This is not a reason to discard the representation theorem, but it is a reason not to install the audit bridge as a default optimization.

13 Further research questions and prioritized tasks

13.1 Native producers before expansion

The first empirical task is to instrument actual scan stages and record candidate template decompositions before object allocation. Can a stage emit registers directly, rather than discover repeated rows after expansion? A useful experiment must include production cost, certificate cost, failed attempts, and peak storage. A small port count measured only after exponential preprocessing is not progress toward the target bound.

13.2 Choosing the base, not just minimizing its ports

Theorem 6.2 solves the optimal port count for a fixed A . The next mathematical problem is to choose a compact template base that simultaneously has small homology, small update rank, and a short coordinate description. Corollary 8.3 supplies a necessary condition for unknot presentations. Can suitable bases be obtained from certified local cancellation families rather than from general matrix optimization?

13.3 Relative, typed ports at tangle frontiers

Can ports be defined in the relevant additive cobordism category so that continuation preserves a compressed representation? The required theorem must identify which basis changes respect object types, which contractions remain valid after gluing, and how coefficient size enters the cost. Merely applying the scalar formula to a residue complex loses information.

13.4 Absorbing tensor amplification

Theorem 9.1 proves that naive scalar-port propagation fails under passive multiplicity amplification. Investigate structured matrix-valued channels or recursive factorizations that can absorb $\Delta \otimes I$ without q separate scalar ports. The central question is whether their singular interactions still admit exact small-core elimination with a proved size bound.

13.5 Graded domains and simultaneous coordinate minimization

The deterministic-language extension represents exact Hamming-weight slices, but a practical constructor may create more complicated degree conditions. Determine which conditions admit small deterministic presentations and whether output registers can be minimized jointly with those conditions. Avoid implicitly replacing a small nondeterministic object by an exponentially large deterministic one.

13.6 A diagram-level constructive parameter theorem

Find a geometric or combinatorial parameter whose bounded value guarantees an efficiently obtainable template-and-port presentation, with explicit bounds for d_ν, B_ν, r , and the number of stages.

Theorem 7.2 then turns that theorem into a runtime result. The parameter must be computable, certified, or supplied with the input; existentially optimal scan orders alone do not satisfy the contract.

13.7 Adversarial lower bounds for register width

Port rank and automaton width are independent sources of cost. Construct families whose updates have small rank but whose coordinate functions require large widths for natural variable orders. Conversely, identify geometric transformations that change the variable order without expanding the complex. Such lower bounds would clarify which representation failures are intrinsic and which are artifacts of a poor layout.

13.8 Composing with the existing restricted algorithms

The twist, graded-occupancy, extremal-window, and radical-transfer methods can produce manageable local complexes in regimes where generic scanning is expensive. Study whether their interfaces give structured low-rank couplings between blocks, rather than multiplying object counts. Every composition theorem must count producer cost and preserve the exact hypotheses of each local result, especially common-disk assumptions and window guard degrees.

13.9 Independent verification and formalization

The replay checker currently shares its arithmetic implementation with the solver. A smaller independent checker could verify rank factorizations, reachability closure, witness evaluations, small pairings, and the rank minor identities. Formalization should begin with Theorem 3.2 and the rank-minimization theorem, then establish the automaton and language semantics. Topological provenance is a separate formalization layer, not an automatic consequence of binary matrix verification.

13.10 Production acceptance criteria

Before a default change, run the full maintained suite, production PD examples, several scan orders, and realistic hard-knot families. Measure failed conversion overhead and capped fallback behavior. Only then investigate a Rust port or more elaborate dense kernels. The present positive abstract cases and negative knot controls define an honest starting point for that work rather than a substitute for it.

14 Conclusion

The main completed advance is an exact polynomial-in-description algorithm for homology of finite-state, port-coupled repeated templates. It remains valid at singular interactions, checks the whole differential, supports regular-language multiplicities, and removes every redundant scalar port for a fixed base without increasing register width. These are executable and mathematically specified operations, not an appeal to an unspecified compression oracle.

The limits are equally concrete. A useful unknot base must already be nearly contractible relative to its port budget; tensoring can force exponential port amplification; open-frontier continuation needs typed relative data; and the available explicit bridge is slower on the measured small knot cubes. The next breakthrough needed for this route is therefore a constructive theorem about

producing and maintaining suitable representations. The consumer-side algebra and its exact cost accounting are provided here as a platform for proving such a theorem.

A Independent small-cube construction

The audit cube represents a braid with s strands and n crossings by $s(n+1)$ level vertices. Unaffected strands are joined vertically. At an active crossing, a smoothing either joins the two strands vertically or joins the two upper endpoints and the two lower endpoints. For positive crossings the two choices are assigned cube coordinates zero and one in that order; for negative crossings the order is reversed. Closure joins the first and last level at each strand. A union-find computation produces the smoothing circles.

Each circle is labeled by 1 or x in $\mathbb{F}_2[x]/(x^2)$. The cube differential uses the standard Frobenius maps

$$m(1,1) = 1, \quad m(1,x) = m(x,1) = x, \quad m(x,x) = 0,$$

$$\Delta(1) = 1 \otimes x + x \otimes 1, \quad \Delta(x) = x \otimes x.$$

An edge of the cube changes the number of circles by one. Comparing the circle partitions identifies whether it is a merge or a split and which labels are unaffected. The homological degree is the number of one resolutions minus the number of negative crossings. Sign assignments are unnecessary over $\mathbb{F}_2$. The standard Frobenius identities make square faces commute; their two paths cancel in characteristic two, giving $d^2 = 0$. The script additionally computes and checks that square directly.

The oracle has strict crossing and dimension caps. It is deliberately exponential and independent of the register representation. Its purpose is to validate small closed complexes and the explicit bridge, not to provide an alternative scalable recognizer. The supplied figure-eight example records the exact braid word, coefficient field, grading convention, constructor, and resulting total rank ten.

B File formats, replay, and reproducibility

The matrix convention is fixed throughout: an integer row has its column zero coefficient in the least significant bit. The JSON format `portkh-1` stores template differential rows, template degrees, port degrees, and the complete register matrices. At one template word, output bits are grouped by template coordinate: first dr bits for the rows of U , followed by dr bits for the columns of V . Thus a V column is stored as an r -bit word, avoiding ambiguity about matrix orientation.

A language descriptor supplies its layer widths, deterministic successor arrays for both symbols, initial state, and accepting states. Language files are arrays aligned with the template blocks; a null entry means all words. The optional language wrapper masks ports and removes the known uncoupled complement exactly, as proved in Proposition 5.1.

From the package root, no dependency installation is required:

```
export PYTHONPATH="$PWD/src"
python -m unittest discover -s tests -v
python scripts/validate.py
python scripts/benchmark.py
python scripts/make_tables.py
sh build.sh
```

The basic large certificate can be replayed by

```
python -m portkh verify \
  examples/connected_singular_40.json \
  examples/connected_singular_40.certificate.json
```

For the language-restricted singular example:

```
python -m portkh verify \
  examples/binomial_63_31.json \
  examples/binomial_63_31.certificate.json \
  --languages examples/binomial_63_31.languages.json
```

The `minimize` command writes a new basic presentation with the minimum homogeneous port count for its fixed base. It does not accept language files directly; a program using language restrictions should apply the proven mask first and keep the accepted-domain bookkeeping. No numerical randomization occurs in the solver. Randomness is used only for reproducible tests and randomized benchmark order.

The archive includes the full source, example inputs and certificates, raw validation and timing records, integration notes, an MIT-0 license, a source-provenance manifest, and file checksums. The PDF and its TeX are both included. Compiling the article uses the adjacent generated `measurements.tex`. Re-running benchmarks changes timing-dependent files; the reproduction script regenerates tables and checksums afterward. The measured numbers in the text describe the packaged run, not a promise that another host will reproduce identical timings.

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- [10] ProveIt, `Topology/UnknotRecognition/synthesis/structural.tex`. Inspected blob prefix 26b056b9a6f8.
- [11] ProveIt, `Topology/UnknotRecognition/synthesis/radical.tex`. Inspected blob prefix a163430fd6cb.